



Class: Final Year B.Tech.

Course: Heating ventilation and air conditioning

Course Code: PEC-ME415

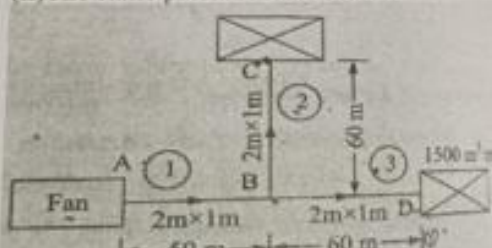
Date: 10/10/2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- All questions are compulsory.

Q.N.		Marks	CO	BT
Q.1	Define Cooling Load and Heating Load.	4	CO3	BT1
Q.2	Compare and contrast Product Load and Process Load in terms of their sources, characteristics, and impact on HVAC system design.	4	CO3	BT1,3,4
Q.3	Explain in detail the Principles of Air Distribution and describe any two types of air distribution systems with neat sketches.	6	CO2	BT1,2
Q.4	An air duct system is provided as shown in the figure below. The air is discharged at point D at a rate of 1500 cubic meters per minute. The friction factor for all ducts is taken as 0.0048. Calculate: (i) The quantity of air discharged at C. (ii) The static pressure at fan outlet. 	6	CO2	BT4

— ALL THE BEST —

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply the concepts of psychometrics and thermodynamics to heating and cooling analysis.
CO 2	Use basic concepts from heat transfer to determine heat gained or lost from a building.
CO 3	Design air-handling systems using concepts from fluid dynamics.
CO 4	Apply good engineering principles to meet the requirements for air quality control and comfort conditions.

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded	
	ISE-II (Semester-I) 2025-26	
	Final. Year. B. Tech.	Mechanical Engineering Department
	Renewable Energy	PEC-ME414
Date: 9/10/25	Time: 14:30 to 15:15	Max. Marks :20

- Note:**
- All questions are compulsory.
 - Assume suitable data if necessary.
 - Use of non-programmable calculator is permitted.
 - This paper contains ONE page.

Q.N.	Question	Marks	CO	BT
Q.1	A solar water heater has a collector area of 3 m^2 and receives solar radiation of 700 W/m^2 . The collector's absorptivity is 0.95, and the transmittance of its glass cover is 0.9. The collector heats 200 kg of water. Assuming heat losses are negligible, and the specific heat of water is $C_p = 4.18 \text{ KJ/kg} \cdot \text{K}$, Estimate the rise in water temperature after 2 hours.	5	CO2	5
Q.2	Explain working of Pyranometer with neat sketch?	5	CO2	2
Q.3	Compare an evacuated tube collector with that of a flat plate collector under identical operating conditions.	5	CO2	2
Q-4	A parabolic dish collector has an aperture diameter of 4 meters and a receiver diameter of 0.2 meters. The average solar irradiance on the dish surface is 900 W/m^2 , and the collector delivers a useful thermal power output of 10 kW. Analyze the system performance by calculating: The Concentration Ratio (CR) of the collector. The Thermal Efficiency (η) of the system.	5	CO2	4

Course Outcomes (CO)

- CO 1: Interpret energy reserves of India and potential of different energy sources.
CO 2: Measure the solar radiation parameters and performance of different solar collectors.
CO 3: Calculate different parameters of wind turbine rotor.
CO 4: Appraise the importance and applications of geothermal and ocean energy.
CO 5: Demonstrate knowledge in field of fuel cell and potential for power generation

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.
In Sem Evaluation II 2025-26
Class: Final Year (Mechanical Engineering)

Date: 09/10/2025
Time: 14:30PM To 15:15PM

Subject: Quality & Reliability Engineering (PCC-ME412)

Max. Marks: 20

Use of non-programmable calculator is permitted.

Assume suitable data if necessary.

Stat tables (A1 to A7) and formulae of control charts with my signature are allowed for examination.

Sr. No.	Questions	M	CO	BT
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- Q.1.** The thickness of the magnetic coating on audio tapes is an important characteristic. (10) CO 3
Random samples of size 4 are selected, and the thickness is measured using an optical instrument. Table shows the mean $\bar{X}$ and standard deviation s for 20 samples. The specifications are 38 ± 4.5 micrometres. If a coating thickness is less than the specifications call for, that tape can be used for a different purpose by running it through another coating operation. 3,4

Sample	Sample Mean, $\bar{X}$	Sample Standard Deviation, s	Sample	Sample Mean, $\bar{X}$	Sample Standard Deviation, s
1	36.4	4.6	11	36.7	5.3
2	35.8	3.7	12	35.2	3.5
3	37.3	5.2	13	38.8	4.7
4	33.9	4.3	14	39.0	5.6
5	37.8	4.4	15	35.5	5.0
6	36.1	3.9	16	37.1	4.1
7	38.6	5.0	17	38.3	5.6
8	39.4	6.1	18	39.2	4.8
9	34.4	4.1	19	36.8	4.7
10	39.5	5.8	20	37.7	5.4

- a) Find the trial control limits and construct a respective charts.
b) Assuming the thickness of the coating to be normally distributed, what proportion of the product will not meet specifications?

- Q.2.** The emergency service unit in a hospital has a goal of 3.5 minutes for the waiting time (05) CO 3
of patients before being treated. A random sample of 20 patients is chosen and the sample average waiting time is found to be 2.3 minutes with a sample standard deviation of 0.5 minutes. Find an appropriate process capability index. Comment on the ability of the emergency service unit to meet the desirable goal, assuming normality. 3,4
What are some possible actions to consider?

- Q.3.** A) Distinguish between a nonconformity and a nonconforming item. Give examples of each in the following contexts: a) Financial institution b) Hospital (03) CO4 4
- B) Which type of control chart is most appropriate to monitor the following situations? (02) CO4 1
- a) Number of firemen to have on duty to provide an acceptable level of service (each fire usually requires four firemen; the department has data on the number of calls they receive during the specified period of time).
- b) Performance of data-entry operators (if one or more input errors are made, the file is rendered useless, and a revision is necessary)

Course Outcomes:

CO 1 Interpret the basic principles of probability theory, standard distributions.

CO 2 Understand and solve numerical on confidence interval, test of hypothesis.

CO 3 Solve numerical examples on statistical process control, rules for out of control, average run length, warning limits, probability of false alarm

CO 4 Solve numerical examples on variable and attribute control charts, process capability analysis.

CO 5 Design a single/double sampling plan, construct its OC curve, and estimate ATI, AOQ and other indices, evaluate economics of sampling.

CO 6 Solve numerical on system reliability estimation.

Bloom's Taxonomy (BT):

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

~ALL THE BEST~

“Trust Yourself, You Know More Than You Think You Do”



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded.
Department of Mechanical Engineering

Academic Year: 2025-2026 [SEM. – 1]

In-Semester Examination - I

Class: Final Year, B.Tech.	Subject: E: Composite Materials	Subject Code: PEC-ME408
Date: 11/10/2025	Time: 14:30-15:15	Max. Marks: 20

- NB:
- This paper contains **One** page (03 Questions). Figures to the right indicate full marks. All questions are compulsory.
 - Draw neat & compact sketches wherever necessary. Write your answers to the point, avoid unnecessary explanations.

Q. No.	Questions	Marks	CO	BT
1.	Discuss the role of <i>particulate reinforcement</i> in composites. Explain how the properties of particulate reinforcements, such as particle size, shape, and purity, affect the performance of composites. Mention the commonly used particulate materials used in different types of composites.	7	CO1, 3	BT1, 2
2.	Briefly explain the Autoclave Processing used for the fabrication of Polymer Matrix Hybrid Composites.	7	CO4	BT1, 2
3.	Enlist at least <i>three</i> typical materials generally used as the <i>continuous phase</i> of Metal Matrix Composites. What are their Advantages? What could be the Advantages of using Metal Matrix composites instead of monolithic metals?	6	CO1, 4	BT1, 2

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Mechanical Engineering In-Semester Examination II Academic Year: 2025-26 (Semester – I)	
Class: Final Year B.Tech.	Subject: Introduction to Industry 4.0	Subject Code: PEC-ME-404
Date: 11/10/2025	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.

Q.N.			Marks	CO	BT
Q.1		Solve any TWO .			
	a	Explain the Cyber Physical system with general block diagram.	05	CO3	BT2
	b	Differentiate between Conventional and Collaborative Robots.	05	CO3	BT4
	c	Explain the following Related Disciplines in Industry 4.0 shortly. 1. Data Science, 2. Artificial Intelligence (AI), 3. Machine Learning (ML), 4. Operations Research, 5. Human-Computer Interaction (HCI)	05	CO3	BT2
Q.2		Solve any TWO .			
	a	Explain the importance of data, information, and knowledge in future organizations.	05	CO4	BT2
	b	Explain the Resource Based view (RBV) of a Firm using VRIO Framework.	05	CO4	BT2
	c	Explain the basics of cloud computing. Distinguish how cloud computing specifically enables different aspects of Industry 4.0, such as IoT, smart manufacturing, and data analytics.	05	CO4	BT2, BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remembering, BT2 - Understanding, BT3 - Applying, BT4 - Analyzing, BT5 - Evaluating, BT6 - Creating

Course Outcomes:

	On successful completion of this course, students should be able to:
CO3.	Identify the various systems used in a manufacturing plant and their role in an Industry 4.0 world.
CO4.	Appreciate the power of Cloud Computing in a networked economy.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Mechanical Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination II

Class: Final Year B.Tech.

Course Name: Finite Element Analysis

Course Code: PEC-ME403

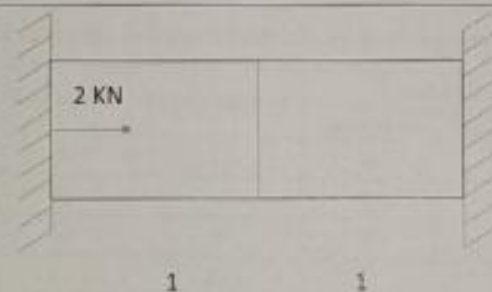
Date: 10-Oct-25

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.
- vi. **Q1 is compulsory solve question 2 or 3**

Q.N.		Marks	CO	BT
Q.1 (a)	Derive shape function for an inclined bar in global coordinate subjected to displacement field of u_1 and u_2 at end nodes. Assume linear variation of field variables.	5	CO2	BT2
Q.1 (b)	Derive a functional of a fin only subjected to convection at free end	5	CO3	BT4
Q.2	$\frac{d^2u}{dx^2} + u = x \quad 0 \leq x \leq 1 \quad u(0) = u(1) = 0$ Solve using RAYLEIGH RITZ METHOD	10	CO3	BT2
Q.3	 Find out the equation of nodal displacement using minimum-potential energy approach	10	CO1	BT3

— ALL THE BEST —

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Analyze and design real world components
CO 2	Suggest whether the given solid is safe for the load applied.
CO 3	Select proper methodology for designing and solving a design problem.
CO 4	Work with different software skills sets required for practical problem.



Class: Final Year, B.Tech.	Subject: Operations Research	Subject Code: HMC-ME402
Date: 13/10/2025	Time: 14.30-15.15	Max. Marks: 20

NB:

- I. This paper contains 02 pages (02 Questions). Figures to right indicate full marks. All Questions are Compulsory.
- II. Outline your solution in clear, sequential steps. Assume suitable data wherever necessary and clearly state it.
- III. Probability distribution chart is printed on Page 2.

Q.No.	Questions	Marks	CO	BT																																																					
1.	A. How will you handle the following situations while solving an assignment problem? (03 Marks) I. Unbalanced Problem II. Restrictions on Assignment B. In an Industry, five different machines can do any of the five jobs, with different profit resulting from each assignment. Find the assignment that maximizes the total profit & amount of total profit. (07 Marks) [Hint: Maximization Problem] <table><tr><th colspan="2"></th><th colspan="5"><u>Machines</u></th></tr><tr><th colspan="2"></th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr><tr><th rowspan="5"><u>Jobs</u></th><th>I</th><td>30</td><td>37</td><td>40</td><td>28</td><td>40</td></tr><tr><th>II</th><td>40</td><td>24</td><td>27</td><td>21</td><td>36</td></tr><tr><th>III</th><td>40</td><td>32</td><td>33</td><td>30</td><td>35</td></tr><tr><th>IV</th><td>25</td><td>38</td><td>40</td><td>36</td><td>36</td></tr><tr><th>V</th><td>29</td><td>62</td><td>40</td><td>34</td><td>39</td></tr></table>			<u>Machines</u>							A	B	C	D	E	<u>Jobs</u>	I	30	37	40	28	40	II	40	24	27	21	36	III	40	32	33	30	35	IV	25	38	40	36	36	V	29	62	40	34	39	10	CO1, 2	BT2, 3,4								
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	III	40	32	33	30	35																																																			
	IV	25	38	40	36	36																																																			
	V	29	62	40	34	39																																																			
2.	A. What does a <i>critical path</i> signify? In what ways all its activities are particularly important? (02 Marks) B. A Project has the following activities & other characteristics: <table><tr><th rowspan="2">Activities</th><th rowspan="2">Predecessor</th><th colspan="3">Time Duration (Weeks)</th></tr><tr><th>Optimistic</th><th>Most Likely</th><th>Pessimistic</th></tr><tr><td>A</td><td>-</td><td>4</td><td>7</td><td>16</td></tr><tr><td>B</td><td>-</td><td>1</td><td>5</td><td>15</td></tr><tr><td>C</td><td>A</td><td>6</td><td>12</td><td>30</td></tr><tr><td>D</td><td>A</td><td>2</td><td>5</td><td>8</td></tr><tr><td>E</td><td>C</td><td>5</td><td>11</td><td>17</td></tr><tr><td>F</td><td>D</td><td>3</td><td>6</td><td>15</td></tr><tr><td>G</td><td>B</td><td>3</td><td>9</td><td>27</td></tr><tr><td>H</td><td>E, F</td><td>1</td><td>4</td><td>7</td></tr><tr><td>I</td><td>G</td><td>4</td><td>19</td><td>28</td></tr></table> Draw the Project Network, Identify the <i>critical path</i> and find the <i>expected project completion time</i>. Also find the probability of completion of project within 40 weeks. (08 Marks)	Activities	Predecessor	Time Duration (Weeks)			Optimistic	Most Likely	Pessimistic	A	-	4	7	16	B	-	1	5	15	C	A	6	12	30	D	A	2	5	8	E	C	5	11	17	F	D	3	6	15	G	B	3	9	27	H	E, F	1	4	7	I	G	4	19	28	10	CO2	BT2, 3
Activities	Predecessor			Time Duration (Weeks)																																																					
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A	-	4	7	16																																																					
B	-	1	5	15																																																					
C	A	6	12	30																																																					
D	A	2	5	8																																																					
E	C	5	11	17																																																					
F	D	3	6	15																																																					
G	B	3	9	27																																																					
H	E, F	1	4	7																																																					
I	G	4	19	28																																																					

CO1. Apply various tools and techniques in finding optimal solutions to problems involving limited resources in the form of – Men, Materials and Machinery.

CO2. Apply various scientific methods of – decision making, project management, inventory management and scheduling different shop-floor processes.

CO3. Evaluate the effects of dynamic changes on optimized solutions.

CO4. Formulate & Analyze the models for real life optimization problems.

BT1 – Remember, BT2 – Understand, BT3 – Apply, BT4 – Analyze, BT5 – Evaluate, BT6 – Create.

z	$\psi(z)$	z	$\psi(z)$	z	$\psi(z)$	z	$\psi(z)$
0.00	0.5000	0.65	0.7422	1.30	0.9032	1.95	0.9744
0.01	0.5040	0.66	0.7454	1.31	0.9049	1.96	0.9750
0.02	0.5080	0.67	0.7486	1.32	0.9066	1.97	0.9756
0.03	0.5120	0.68	0.7517	1.33	0.9082	1.98	0.9761
0.04	0.5160	0.69	0.7549	1.34	0.9099	1.99	0.9767
0.05	0.5199	0.70	0.7580	1.35	0.9115	2.00	0.9772
0.06	0.5239	0.71	0.7611	1.36	0.9131	2.02	0.9783
0.07	0.5279	0.72	0.7642	1.37	0.9147	2.04	0.9793
0.08	0.5319	0.73	0.7673	1.38	0.9162	2.06	0.9803
0.09	0.5359	0.74	0.7703	1.39	0.9177	2.08	0.9812
0.10	0.5398	0.75	0.7734	1.40	0.9192	2.10	0.9821
0.11	0.5438	0.76	0.7764	1.41	0.9207	2.12	0.9830
0.12	0.5478	0.77	0.7794	1.42	0.9222	2.14	0.9838
0.13	0.5517	0.78	0.7823	1.43	0.9236	2.16	0.9846
0.14	0.5557	0.79	0.7852	1.44	0.9251	2.18	0.9854
0.15	0.5596	0.80	0.7881	1.45	0.9265	2.20	0.9861
0.16	0.5636	0.81	0.7910	1.46	0.9279	2.22	0.9868
0.17	0.5675	0.82	0.7939	1.47	0.9292	2.24	0.9875
0.18	0.5714	0.83	0.7967	1.48	0.9306	2.26	0.9881
0.19	0.5753	0.84	0.7995	1.49	0.9319	2.28	0.9887
0.20	0.5793	0.85	0.8023	1.50	0.9332	2.30	0.9893
0.21	0.5832	0.86	0.8051	1.51	0.9345	2.32	0.9898
0.22	0.5871	0.87	0.8078	1.52	0.9357	2.34	0.9904
0.23	0.5910	0.88	0.8106	1.53	0.9370	2.36	0.9909
0.24	0.5948	0.89	0.8133	1.54	0.9382	2.38	0.9913
0.25	0.5987	0.90	0.8159	1.55	0.9394	2.40	0.9918
0.26	0.6026	0.91	0.8186	1.56	0.9406	2.42	0.9922
0.27	0.6064	0.92	0.8212	1.57	0.9418	2.44	0.9927
0.28	0.6103	0.93	0.8238	1.58	0.9429	2.46	0.9931
0.29	0.6141	0.94	0.8264	1.59	0.9441	2.48	0.9934
0.30	0.6179	0.95	0.8289	1.60	0.9452	2.50	0.9938
0.31	0.6217	0.96	0.8315	1.61	0.9463	2.52	0.9941
0.32	0.6255	0.97	0.8340	1.62	0.9474	2.54	0.9945
0.33	0.6293	0.98	0.8365	1.63	0.9484	2.56	0.9948
0.34	0.6331	0.99	0.8389	1.64	0.9495	2.58	0.9951
0.35	0.6368	1.00	0.8413	1.65	0.9505	2.60	0.9953
0.36	0.6406	1.01	0.8438	1.66	0.9515	2.62	0.9956
0.37	0.6443	1.02	0.8461	1.67	0.9525	2.64	0.9959
0.38	0.6480	1.03	0.8485	1.68	0.9535	2.66	0.9961
0.39	0.6517	1.04	0.8508	1.69	0.9545	2.68	0.9963
0.40	0.6554	1.05	0.8531	1.70	0.9554	2.70	0.9965
0.41	0.6591	1.06	0.8554	1.71	0.9564	2.72	0.9967
0.42	0.6628	1.07	0.8577	1.72	0.9573	2.74	0.9969
0.43	0.6664	1.08	0.8599	1.73	0.9582	2.76	0.9971
0.44	0.6700	1.09	0.8621	1.74	0.9591	2.78	0.9973
0.45	0.6736	1.10	0.8643	1.75	0.9599	2.80	0.9974
0.46	0.6772	1.11	0.8665	1.76	0.9608	2.82	0.9976
0.47	0.6808	1.12	0.8686	1.77	0.9616	2.84	0.9977
0.48	0.6844	1.13	0.8708	1.78	0.9625	2.86	0.9979
0.49	0.6879	1.14	0.8729	1.79	0.9633	2.88	0.9980
0.50	0.6915	1.15	0.8749	1.80	0.9641	2.90	0.9981
0.51	0.6950	1.16	0.8770	1.81	0.9649	2.92	0.9982
0.52	0.6985	1.17	0.8790	1.82	0.9656	2.94	0.9984
0.53	0.7019	1.18	0.8810	1.83	0.9664	2.96	0.9985
0.54	0.7054	1.19	0.8830	1.84	0.9671	2.98	0.9986
0.55	0.7088	1.20	0.8849	1.85	0.9678	3.00	0.99865
0.56	0.7123	1.21	0.8869	1.86	0.9686	3.20	0.99931
0.57	0.7157	1.22	0.8888	1.87	0.9693	3.40	0.99966
0.58	0.7190	1.23	0.8907	1.88	0.9699	3.60	0.999841
0.59	0.7224	1.24	0.8925	1.89	0.9706	3.80	0.999928
0.60	0.7257	1.25	0.8944	1.90	0.9713	4.00	0.999968
0.61	0.7291	1.26	0.8962	1.91	0.9719	4.50	0.999997
0.62	0.7324	1.27	0.8980	1.92	0.9726	5.00	0.999997
0.63	0.7357	1.28	0.8997	1.93	0.9732		
0.64	0.7389	1.29	0.9015	1.94	0.9738		



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In Semester Examination - I
Academic Year: 2025-26 (Semester – I)

Class: TY	Subject: Modern Mobility/Internal Combustion Engines and E-Vehicles	Subject Code: PECME305/PEC-ME-311
Date: 13.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. The symbols carry the usual meaning.
- iii. This paper contains 1 page

Q.N.		Marks	CO	BT
1.	Illustrate the four stages of combustion in an SI engine on a P- θ (pressure–crank angle) diagram and describe significance of each stage in engine operation.	5	2	2
2.	Construct a tabular layout of a typical heat balance sheet. Explain how a heat balance sheet helps in improving the performance of an IC engine.	5	3	3
3.	A gasoline engine working on four stroke develops a brake power of 20.9 kW. A Morse Test was conducted on this engine and the brake power (kW) obtained when each cylinder was made inoperative by short circuiting the spark plug is 14.9, 14.3, 14.8 and 14.5 respectively. The test was conducted at constant speed. Evaluate the indicated power and friction power for each cylinder.	5	3	5
4.	Explain how a catalytic converter helps in reducing exhaust emissions.	5	3	2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO1. Analyse IC engine components, working, and cycles CO2. Understand combustion processes and ignition systems in engines CO3. Evaluate engine performance and apply emission control strategies CO4. Assess electric vehicle technologies and global trends CO5. Optimize EV performance through motor and battery management CO6. Design hybrid powertrains and explore sustainable transportation

			Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Mechanical Engineering In-Semester Examination I Academic Year: 2025-26 (Semester I)		
Class: TY		Subject: Manufacturing Processes - II		Subject Code: PCC-ME304	
Date: 11- Oct-2025		Time: 11:00 to 11:45		Max. Marks: 20	

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Solve all the questions	Marks	CO	BT
Q.1	Explain the types of Milling cutters with a diagram.	5	CO4	BT1
Q.2	Elaborate on the Broaching tool in detail, with diagrams, advantages and disadvantages and application	5	CO2	BT1
Q.3	How the lathe machine can be divided into various groups?	5	CO4	BT2
Q.4	In a mild steel block, a flat surface of length 100 mm and width 60 mm has to be finished in a shaping machine in a single pass. How much machining time will be required if $N_s = 80$, $s_o = 0.2$ mm/stroke, $A = O = 5$ mm, $QRR = 0.5$.	5	CO3	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Demonstrate the metal cutting process and select the appropriate cutting tools for it
CO 2	Exhibit the characteristic, terminologies related to cutting tools.
CO 3	Calculate the machining time, depth of cut, feed required.
CO 4	Demonstrate knowledge related to various machining processes.
CO 5	Exhibits the knowledge of various finishing processes.
CO 6	Select proper machining processes for the product to be manufacture



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

In Semester Examination (Semester-I) 2025-26

T.Y.B. Tech.

Introduction to Industry 4.0

Mechanical Engineering

PEC-ME303

Date: 09-OCT-25 Time: 11:00-11:45

Max. Marks: 20

- Note: i. Assume suitable data, if necessary.
ii. Solve **any four questions** from questions 1 to 5.
iii. Sketches, Flowcharts to right indicate full marks.
iv. Paper containing 1 page.

- | | | | | |
|-----|---|---|-----|-------|
| Q.1 | Recently you have seen flood war was everywhere and hence, "Drones equipped with sensor's are deployed during floods to detect survivors". Explain the CPS architecture used for communication, control and coordination. | 5 | CO3 | BT3,2 |
| Q.2 | Explain And draw neat sketch of four configurations of Robots. | 5 | CO5 | BT2 |
| Q.3 | Explain the concept of Mobile computing and provide examples of their applications in modern industries. | 5 | CO4 | BT2 |
| Q.4 | What are the potential cyber security challenges in the implementation of cyber physical system? Discuss the measures required to ensure the security of these systems. | 5 | CO6 | BT1,2 |
| Q.5 | Define cloud computing and describe its basic components. How does cloud computing enable organizations to scale their operations and improve flexibility? | 5 | CO4 | BT1,2 |

Course Outcomes (CO)

- CO1. Understand the drivers and enablers of Industry 4.0.
CO2. Appreciate the smartness in Smart Factories, Smart, Cities, Smart Products, and Smart Services.
CO3. Identify the various systems used in a manufacturing plant and their role in an Industry 4.0 world.
CO4. Appreciate the power of cloud computing in a networked economy.
CO5. Understand the opportunities and challenges brought about by Industry 4.0.
CO6. Analyze how organizations and individuals should prepare to reap the benefits of Industry 4.0.

Bloom's Taxonomy:BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

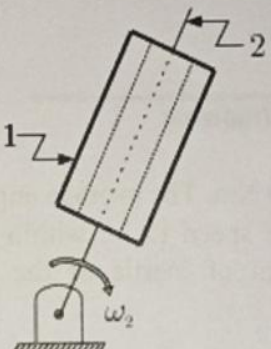
~~ Best Wishes ~~

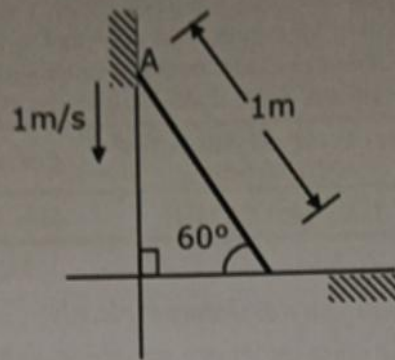


Class: TY B.Tech.	Course: Theory of Machines and Mechanisms	Course Code: PCC-ME303
Date: 11.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

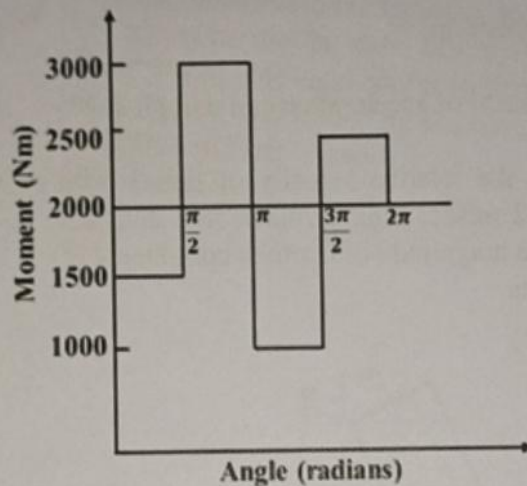
Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	a Explain Arnhold Kennedy's theorem.	03	CO2	BT2
	b 1) When Coriolis component of acceleration come in picture?	01	CO2	BT1
	2) In the figure shown, the relative velocity of link 1 with respect to link 2 is 12 m/sec. Link 2 rotates at a constant speed of 120 rpm. The magnitude of Coriolis component of acceleration of link 1 is	02		BT3
				
	c The rotor of a turbojet engine of an aircraft has a mass 180 kg and polar moment of inertia $10 \text{ kg} \cdot \text{m}^2$ about the rotor axis. The rotor rotates at a constant speed of 1100 rad/s in the clockwise direction when viewed from the rear of the aircraft. The aircraft while flying at a speed of 800 km per hour takes a turn with a radius of 1.5 km to the right. The gyroscopic moment exerted by the rotor on the aircraft structure and the direction of motion of the nose when the aircraft turns, are	04	CO4	BT3
	d A rod of length 1 m is sliding in a corner as shown in Fig. At an instant when the rod makes an angle of 60 degrees with the horizontal plane, the velocity of point A on the rod is 1 m/s. The angular velocity of the rod at this instant.	02	CO2	BT1

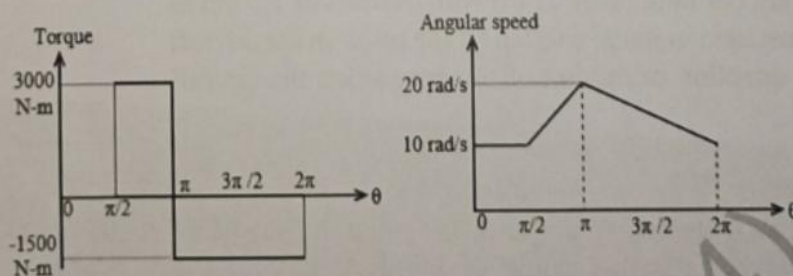


- Q.2 a If C is the coefficient of speed fluctuation of a flywheel, then the ratio of $\omega_{max}/\omega_{min}$ in terms of C will be: 02 CO4 BT3
- b The turning moment diagram of a flywheel fitted to a fictitious engine is shown in the figure. 03 CO4 BT13



The mean turning moment is 2000 Nm. The average engine speed is 1000 rpm. For fluctuation in the speed to be within $\pm 2\%$ of the average speed, the mass moment of inertia of the flywheel is _____ kgm^2 .

- c Torque and angular speed data over one cycle for a shaft carrying a flywheel are shown in the figures. The moment of inertia (in $kg.m^2$) of the flywheel is _____ 03 CO4 BT2, BT3





Class: TY B.Tech.

Course Name: Dynamics of
Machines

Course Code: PCC-ME302

Date: 09.10.2025

Time: 11:00 to 11:45

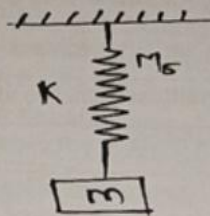
Max. Marks: 20

Note:

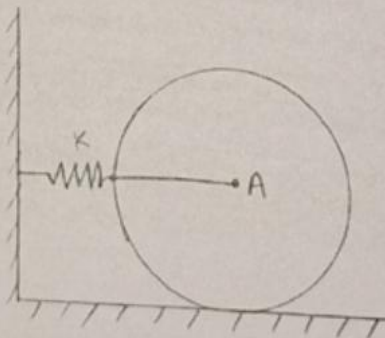
- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.

- | | | Marks | CO | BT |
|-----|---|-------|-----|-----|
| Q.1 | a | 05 | CO5 | BT3 |
- Derive the natural frequency of spring mass system with spring having mass M_s for given system.



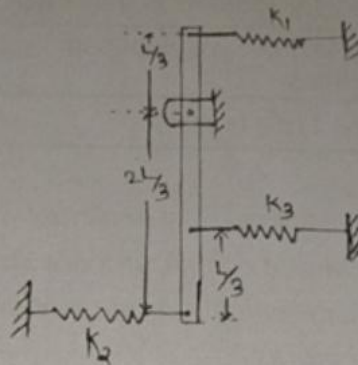
- b A uniform thin disk of mass 1 kg and radius 0.1 m is kept on a surface as shown in the figure. A spring of stiffness $k = 400 \text{ N/m}$ is connected to the disk centre A. Initially, the springs are unstretched. Assume pure rolling of the disk. For small disturbance from the equilibrium, the natural frequency of vibration of the system is _____ rad/s (round off to one decimal place)



- | | | | | |
|-----|---|----|-----|-------------|
| Q.2 | a | 05 | CO3 | BT1,
BT2 |
|-----|---|----|-----|-------------|
- State D'Alembert's Principle. Explain it.

- b Find the natural frequencies of given vibrations system as rod having mass m and total length L . Spring stiffness K_1 , K_2 and K_3 as shown in fig.

05 CO5 BT2, BT3



=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Ability to analyze the gear trains used in various machines.
CO 2	Student demonstrates and analyzes different mechanisms used in various machines.
CO 3	Student exhibit skills towards application of static force analysis and synthesis of mechanisms.
CO 4	Student exhibits need, design and applications of balancing machine parts.
CO 5	Student exhibits solutions to issues related to vibration, their effect and its control measures.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester-I)

Class: TY	Subject: Heat Transfer	Subject Code: PCC-ME-301
Date: 10.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- This paper contains one page.

Q.N.		Marks	CO	BT
Q.1	A turbine blade made of a metal alloy ($k=17 \text{ W/m}\cdot\text{K}$) has a length of 5.3 cm, a perimeter of 11 cm, and a cross-sectional area of 5.13 cm^2. The turbine blade is exposed to hot gas from the combustion chamber at 973°C with a convection heat transfer coefficient of $538 \text{ W/m}^2\cdot\text{K}$. The base of the turbine blade maintains a constant temperature of 450°C and the tip is adiabatic. Utilize appropriate fin analysis to determine the heat transfer rate to the blade and the temperature at the tip. T_t Hot gas, 973°C $h = 538 \text{ W/m}^2\cdot\text{K}$ Turbine blade $k = 17 \text{ W/m}\cdot\text{K}$ $p = 11 \text{ cm}$, $L = 5.3 \text{ cm}$ $A_c = 5.13 \text{ cm}^2$ $T_b = 450^\circ\text{C}$	6	CO 1-3	BT 3-4
Q.2	Two infinite parallel plates are placed at a certain distance apart. An infinite radiation shield is inserted between the plates without touching any of them to reduce heat exchange between the plates. Assume that the emissivities of plates and radiation shield are equal. Derive the ratio of the net heat exchange between the plates with and without the shield.	4	CO2	BT 2-4
Q.3	A solid sphere of radius 10 mm is placed at the centroid of a hollow cubical enclosure of side length 30 mm. The outer surface of the sphere is denoted by 1 and the inner surface of the cube is denoted by 2. Utilize the relationship between view factors to solve for the view factor F_{22} governing radiative heat transfer between the surfaces.	4	CO2	BT 3-4
Q.4	Water ($C_p = 4180 \text{ J/kg}\cdot\text{K}$) is to be heated by hot air ($C_p = 1010 \text{ J/kg}\cdot\text{K}$) in a thin-walled double pipe counter flow heat exchanger. Air having the mass flow rate of 0.4 kg/s enters at 90°C while water having the mass flow rate of 0.1 kg/s enters at 22°C . The inner diameter and length of tubes are 1.2 cm and 12 m respectively. The overall heat transfer coefficient based on inner diameter is $80 \text{ W/m}^2\cdot\text{K}$. Find out (a) Effectiveness of heat exchanger. (b) Heat transfer rate (c) Temperature of water at outlet.	6	CO 3-4	BT 3-4

— ALL THE BEST —

Bloom's Taxonomy (BT) BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Student will be able to analyze heat conduction with and without internal heat generation, critical thickness of insulation and extended surfaces with the practical utilities.
CO 2	Student demonstrates the concept and mechanism of convection, conduction and radiation.
CO 3	Student exhibits application of mathematical approach to analyze and solve the numerical on complex heat transfer phenomena's.
CO 4	Student can explain the design, performance analysis and practical applications of heat exchangers.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

ISE II (Semester-2) 2025-26

S.Y. B. Tech.

Strength of materials

Mechanical Engineering

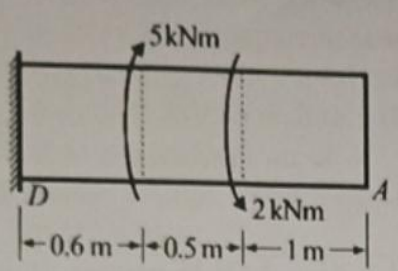
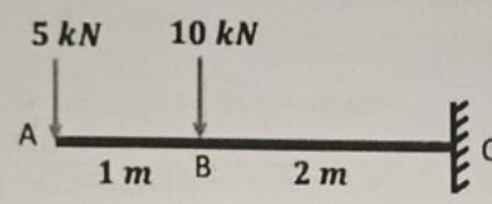
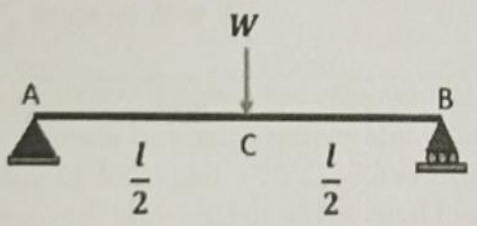
PCC-ME201

Date: 10/10/2025

Time: 14:30 to 15:15

Max. Marks: 20

Note: All questions are compulsory

Q.N.	Question	Marks	CO	BT
Q.1	A shaft is subjected to torque as shown in figure. Diameter of the shaft is 40 mm and the modulus of rigidity is 80 GPa. Find the maximum shear stress in MPa and the angle of twist at A in degrees. 	5	CO4	BT4
Q. 2	Find the deflection at the end A (in mm) of the beam shown in figure. Assume flexural rigidity = 10^4 kNm². 	5	CO4	BT2
Q. 3	A solid circular shaft of diameter 50 mm is subjected to a torque of 1600 N-m. $G = 80$ GPa. Find the maximum shear stress (in MPa) and angle of twist (in degree) in a length of 1 m.	5	CO3	BT3
Q. 4	Find the maximum slope and deflection of the beam shown in figure. Assume flexural rigidity = EI. 	5	CO4	BT3

Course Outcomes (CO)

- CO1. Understand the various concepts and principles used in mechanics of deformable materials.
- CO2. Evaluate the strains and deformation that will result due to the elastic stresses developed within the materials for simple types of loading.
- CO3. Analyze various types of stress, strains and deformations in beams, shafts, pressure vessels, etc.
- CO4. Evaluate the slope, deflections in beams and columns.
- CO5. Analyze the failure of columns under various loadings.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



General Instructions:

- I. This paper contains 1 + 2 = 3 pages. Figures to right indicate full marks.
- II. Draw neat & compact sketches wherever necessary.
- III. Assume data wherever required and state them clearly.

QN.	Question	Marks	CO	BT
1.	Last year in the summer, department faced the water crises due to lower capacity of the water cooler. In this regard, department is planning to increase the cooling capacity of water cooler. Based on the expected usage, it is required to produce a refrigerating effect of 15 kW to maintain the cold water temperature at 20° C by rejecting heat to the ambient air at 37° C. Considering the water cooler is operating as Reversed Carnot Cycle, help the department identifying the power required to run the system.	5	CO5	BT5
2.	Consider a container with volume of 80 L with 4 kg of water at pressure of 1.5bar. For this condition, Determine – a. Temperature b. Quality c. Enthalpy d. Volume of vapor e. Volume of liquid. <i>Hint: Use the steam-table provided as an attachment. [No need to attach the table with answer sheets.]</i>	5	CO3	BT5
3.	Identify the relation for below conditions using =, +, –, ×, ÷, <, > . –Compared to saturated vapors, superheated vapors are characterized by: a. Lower pressure: $P _ P_{sat}$ at given T b. Higher temperature: $T _ T_{sat}$ at given P c. Specific Volume: $V _ V_g$ at given P or T d. Internal Energies: $u _ u_g$ at given P or T e. Enthalpy: $h _ h_g$ at given P or T .	5	CO3	BT3
4.	A reversible heat engine operates between two reservoirs at temperature of 600°C and 40°C. The engine drives a reversible refrigerator which operates between reservoirs at temperature of 40°C and –20°C. The heat transfer to the heat engine is 2000 kJ and the net work output of the combined engine refrigerator plant is 360 kJ. Evaluate the heat transfer to the refrigerant and the net heat transfer to the reservoir at 40°C.	5	CO5	BT5

**** ALL THE BEST ****

CO1: To impart the principles of work and energy.

CO2: To inculcate fundamentals of thermodynamics laws, concepts & principles.

CO3: To comprehend knowledge about properties of pure substance.

CO4: To comprehend knowledge about entropy, irreversibility and availability

CO5: To make familiar with various thermodynamic cycles and to apply thermodynamic concepts in various applications.

BT1: Remember, **BT2:** Understand, **BT3:** Apply, **BT4:** Analyze, **BT5:** Evaluate, **BT6:** Create.

Water/Steam at $p = 0.15 \text{ MPa}$

Water/Steam at $p = 0.15 \text{ MPa}$					Water/Steam at $p = 0.15 \text{ MPa}$				
T	v	u	h	s	T	v	u	h	s
$^{\circ}\text{C}$	m^3/kg	kJ/kg	kJ/kg	kJ/kg K	$^{\circ}\text{C}$	m^3/kg	kJ/kg	kJ/kg	kJ/kg K
0	0.00100013	-0.04	0.11	-0.00014	270	1.6637	2763.3	3012.9	7.9202
5	0.00100001	21.02	21.17	0.07625	280	1.6949	2778.8	3033.0	7.9569
10	0.00100027	42.02	42.17	0.15107	290	1.7260	2794.2	3053.1	7.9929
15	0.00100088	62.92	63.12	0.22444	300	1.7571	2809.7	3073.3	8.0284
20	0.00100177	83.90	84.05	0.29645	310	1.7882	2825.3	3093.5	8.0634
25	0.00100294	104.82	104.97	0.36719	320	1.8192	2840.9	3113.8	8.0978
30	0.00100435	125.72	125.87	0.43671	330	1.8503	2856.6	3134.1	8.1318
35	0.00100598	146.61	146.76	0.50508	340	1.8813	2872.3	3154.5	8.1653
40	0.00100782	167.51	167.66	0.57235	350	1.9123	2888.1	3174.9	8.1983
45	0.00100986	188.41	188.56	0.63855	360	1.9433	2903.8	3195.3	8.2309
50	0.00101209	209.31	209.46	0.70374	370	1.9743	2919.8	3215.9	8.2631
55	0.00101449	230.22	230.37	0.76796	380	2.0052	2935.6	3236.4	8.2948
60	0.00101707	251.14	251.29	0.83122	390	2.0362	2951.7	3257.1	8.3262
65	0.00101981	272.07	272.22	0.89358	400	2.0671	2967.7	3277.8	8.3572
70	0.00102272	293.01	293.16	0.95506	410	2.0981	2983.8	3298.5	8.3878
75	0.00102578	313.97	314.12	1.0157	420	2.1290	3000.0	3319.4	8.4180
80	0.00102901	334.94	335.09	1.0755	430	2.1599	3016.3	3340.3	8.4480
85	0.00103238	355.94	356.09	1.1345	440	2.1908	3032.6	3361.2	8.4775
90	0.00103591	376.94	377.10	1.1928	450	2.2217	3048.9	3382.2	8.5068
95	0.00103960	397.98	398.14	1.2503	460	2.2526	3065.4	3403.3	8.5357
100	0.00104344	419.04	419.20	1.3072	470	2.2835	3081.9	3424.4	8.5644
105	0.00104743	440.14	440.30	1.3633	480	2.3144	3098.4	3445.6	8.5927
110	0.00105158	461.26	461.42	1.4188	490	2.3453	3115.1	3466.9	8.6207
111.349	0.00105273	466.97	467.13	1.4337	500	2.3761	3131.8	3488.2	8.6485
111.349	1.1593	2519.2	2693.1	7.2230	520	2.4379	3165.4	3531.1	8.7032
115	1.1714	2525.1	2700.8	7.2430	540	2.4996	3199.3	3574.2	8.7569
120	1.1880	2533.2	2711.4	7.2699	560	2.5613	3233.4	3617.6	8.8096
125	1.2044	2541.1	2721.8	7.2962	580	2.6230	3267.8	3661.2	8.8614
130	1.2208	2549.0	2732.1	7.3220	600	2.6846	3302.5	3705.2	8.9124
135	1.2370	2556.8	2742.4	7.3473	620	2.7463	3337.5	3749.4	8.9624
140	1.2533	2564.6	2752.6	7.3722	640	2.8080	3372.7	3793.9	9.0117
145	1.2694	2572.4	2762.8	7.3967	660	2.8696	3408.3	3838.7	9.0602
150	1.2855	2580.1	2772.9	7.4208	680	2.9312	3444.0	3883.7	9.1080
155	1.3016	2587.8	2783.0	7.4445	700	2.9929	3480.2	3929.1	9.1550
160	1.3176	2595.5	2793.1	7.4679	720	3.0545	3516.5	3974.7	9.2014
165	1.3335	2603.1	2803.1	7.4910	740	3.1161	3553.2	4020.6	9.2472
170	1.3495	2610.8	2813.2	7.5138	760	3.1777	3590.1	4066.8	9.2923
175	1.3654	2618.4	2823.2	7.5363	780	3.2394	3627.3	4113.2	9.3369
180	1.3813	2626.0	2833.2	7.5585	800	3.3010	3664.9	4160.0	9.3808
185	1.3971	2633.6	2843.2	7.5804	820	3.3626	3702.6	4207.0	9.4243
190	1.4129	2641.3	2853.2	7.6021	840	3.4242	3740.7	4254.3	9.4671
195	1.4287	2648.8	2863.1	7.6235	860	3.4858	3779.0	4301.9	9.5095
200	1.4445	2656.4	2873.1	7.6447	880	3.5473	3817.6	4349.7	9.5513
210	1.4760	2671.6	2893.0	7.6864	900	3.6089	3856.5	4397.8	9.5927
220	1.5074	2686.9	2913.0	7.7272	920	3.6705	3895.6	4446.2	9.6336
230	1.5388	2702.1	2932.9	7.7672	940	3.7321	3935.1	4494.9	9.6740
240	1.5701	2717.4	2952.9	7.8065	960	3.7937	3974.7	4543.8	9.7140
250	1.6013	2732.7	2972.9	7.8451	980	3.8553	4014.7	4593.0	9.7536
260	1.6325	2748.0	2992.9	7.8830	1000	3.9168	4054.9	4642.4	9.7927
270	1.6637	2763.3	3012.9	7.9202					

Saturated Water and Steam (Pressure-based)

p MPa	T_{sat}	Volume, m ³ /kg		Energy, kJ/kg		Enthalpy, kJ/kg			Entropy, kJ/(kg K)		
		v_f	v_g	u_f	u_g	h_f	h_g	h_{fg}	s_f	s_g	s_{fg}
0.050	81.317	0.00102993	3.2400	340.49	2483.2	340.54	2645.2	2304.7	1.0912	7.5930	6.5018
0.055	83.709	0.00103154	2.9635	350.53	2486.2	350.59	2649.2	2298.6	1.1194	7.5606	6.4412
0.060	85.926	0.00103307	2.7317	359.85	2489.0	359.91	2652.9	2292.9	1.1454	7.5311	6.3857
0.065	87.993	0.00103452	2.5346	368.53	2491.6	368.60	2656.3	2287.7	1.1696	7.5040	6.3345
0.070	89.932	0.00103590	2.3648	376.68	2493.9	376.75	2659.4	2282.7	1.1921	7.4790	6.2869
0.075	91.758	0.00103723	2.2170	384.36	2496.1	384.44	2662.4	2277.9	1.2132	7.4557	6.2425
0.080	93.486	0.00103850	2.0871	391.63	2498.2	391.71	2665.2	2273.5	1.2330	7.4339	6.2009
0.085	95.125	0.00103972	1.9720	398.53	2500.2	398.62	2667.8	2269.2	1.2518	7.4135	6.1617
0.090	96.687	0.00104091	1.8694	405.11	2502.1	405.20	2670.3	2265.1	1.2696	7.3943	6.1246
0.095	98.178	0.00104205	1.7772	411.38	2503.9	411.48	2672.7	2261.2	1.2866	7.3761	6.0895
0.10	99.606	0.00104315	1.6939	417.40	2505.5	417.50	2674.9	2257.4	1.3028	7.3588	6.0561
0.11	102.292	0.00104527	1.5495	428.73	2508.8	428.84	2679.2	2250.3	1.3330	7.3269	5.9938
0.12	104.784	0.00104727	1.4284	439.23	2511.7	439.36	2683.1	2243.7	1.3609	7.2977	5.9367
0.13	107.109	0.00104917	1.3253	449.05	2514.3	449.19	2686.6	2237.5	1.3868	7.2709	5.8840
0.14	109.292	0.00105099	1.2366	458.27	2516.9	458.42	2690.0	2231.6	1.4110	7.2461	5.8351
0.15	111.349	0.00105273	1.1593	466.97	2519.2	467.13	2693.1	2226.0	1.4337	7.2230	5.7893
0.16	113.297	0.00105440	1.0914	475.21	2521.4	475.38	2696.0	2220.7	1.4551	7.2014	5.7463
0.17	115.148	0.00105600	1.0312	483.04	2523.5	483.22	2698.8	2215.6	1.4753	7.1812	5.7059
0.18	116.911	0.00105756	0.97747	490.51	2525.5	490.70	2701.4	2210.7	1.4945	7.1621	5.6676
0.19	118.596	0.00105906	0.92924	497.65	2527.3	497.85	2703.9	2206.0	1.5127	7.1440	5.6313
0.20	120.210	0.00106052	0.88568	504.49	2529.1	504.70	2706.2	2201.5	1.5302	7.1269	5.5967
0.21	121.759	0.00106193	0.84614	511.07	2530.8	511.29	2708.5	2197.2	1.5469	7.1106	5.5638
0.22	123.250	0.00106330	0.81007	517.40	2532.4	517.63	2710.6	2193.0	1.5628	7.0951	5.5323
0.23	124.686	0.00106464	0.77704	523.50	2534.0	523.74	2712.7	2188.9	1.5782	7.0803	5.5021
0.24	126.072	0.00106594	0.74668	529.38	2535.4	529.64	2714.6	2185.0	1.5930	7.0661	5.4731
0.25	127.411	0.00106722	0.71866	535.07	2536.8	535.34	2716.5	2181.1	1.6072	7.0524	5.4452
0.26	128.708	0.00106846	0.69273	540.59	2538.2	540.87	2718.3	2177.4	1.6210	7.0394	5.4184
0.27	129.965	0.00106968	0.66845	545.95	2539.5	546.24	2720.0	2173.8	1.6343	7.0268	5.3925
0.28	131.185	0.00107086	0.64624	551.14	2540.8	551.44	2721.7	2170.3	1.6471	7.0146	5.3675
0.29	132.370	0.00107203	0.62533	556.19	2542.0	556.50	2723.3	2166.8	1.6596	7.0029	5.3433
0.30	133.522	0.00107317	0.60576	561.11	2543.2	561.43	2724.9	2163.5	1.6717	6.9916	5.3199
0.31	134.644	0.00107429	0.58741	565.89	2544.3	566.22	2726.4	2160.2	1.6835	6.9807	5.2972
0.32	135.737	0.00107539	0.57017	570.56	2545.3	570.90	2727.8	2157.0	1.6949	6.9701	5.2752
0.33	136.802	0.00107647	0.55395	575.10	2546.5	575.46	2729.3	2153.8	1.7060	6.9598	5.2538
0.34	137.842	0.00107753	0.53864	579.54	2547.5	579.91	2730.6	2150.7	1.7168	6.9498	5.2330
0.35	138.857	0.00107857	0.52418	583.88	2548.5	584.26	2732.0	2147.7	1.7274	6.9401	5.2128
0.36	139.849	0.00107960	0.51050	588.13	2549.4	588.52	2733.2	2144.7	1.7377	6.9307	5.1931
0.37	140.819	0.00108061	0.49753	592.28	2550.4	592.68	2734.5	2141.8	1.7477	6.9216	5.1739
0.38	141.769	0.00108161	0.48522	596.34	2551.3	596.75	2735.7	2139.0	1.7575	6.9126	5.1551
0.39	142.698	0.00108259	0.47352	600.32	2552.2	600.74	2736.9	2136.2	1.7671	6.9040	5.1369
0.40	143.608	0.00108355	0.46238	604.22	2553.1	604.65	2738.1	2133.4	1.7765	6.8955	5.1190